

Practical Exam Unit 1

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Introduction

For the exam of “statistics and probability”, we were asked about doing an analysis of a data set, this was interesting to show so the solution that I’ve come from was to select a database from *kaggle.com*, in this case, I have decided to use a database called “*Extrovert vs. Introvert Behavior Data*”, where shows statistics about extroverts and introverts behaviors and characteristics. You can find the data set here: <https://www.kaggle.com/datasets/rakeshkapilavai/extrovert-vs-introvert-behavior-data?resource=download>

Resolution of data

Importation of data

Data must be imported in R in order to use it, for this you must download the “*readxl*” library in order to open the data and align to the environment.

```
#Import
library("readxl")

data <- read_excel("C:/Users/LAB2BPC1/Downloads/personality_dataset.xlsx")
View(data)
data1 <- na.omit(data)
View(data1)
```

This are all the steps used in order to portrait the data.

Qualitative data

Qualitative data are the type of data that describes a non-countable characteristic, such as *names*, *colors*, *personality*, *etc.* In this case, personality (extrovert vs introvert) was used. But before this you must assign the distinct values into a identifier

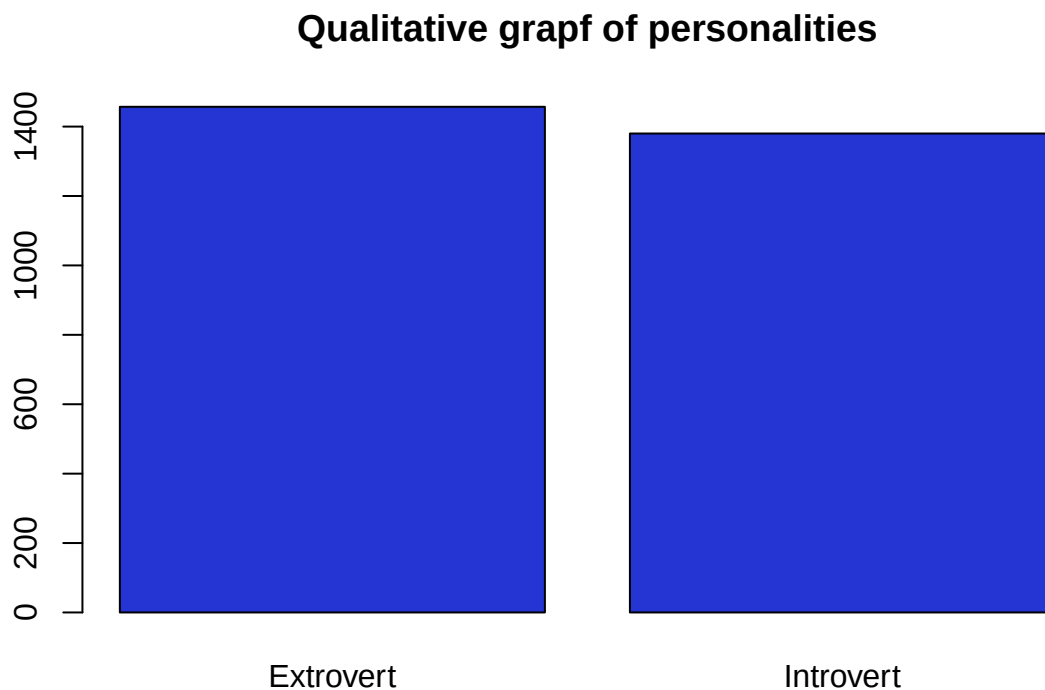
```
#Data frag
personal <- data1$Personality
T_personal <- table(personal)
```

After this, you can see what is the frequency of the qualitative data and a plot to show it more visually.

```
#Qualitative
print(T_personal) #Count of qualitative data

## personal
## Extrovert Introvert
##      1457      1380

barplot(T_personal, col = "#2435D4", main = "Qualitative grapf of personalities") #Bar plot
```



This was everything used in order to show the qualitative data.

Quantitative data

Quantitative data are the type of data that describes a countable characteristic, such as *ages*, *heights*, *times*, *etc.* In this case, time spent alone was used. But before this you must assign the distinct values into a identifier

```
#Data frag
timeAlone <- data1$Time_spent_Alone
T_timeAlone <- table(timeAlone)
```

After this, you can see what is the quantitative statistics of the data set, such as mean, medium, mode, range, variance and standard deviation

```
#Quantitative  
mean(timeAlone) #mean
```

```
## [1] 4.505816
```

```
median(timeAlone) #median
```

```
## [1] 4
```

```
print(T_timeAlone) #mode (in this case "0" repeats more)
```

```
## timeAlone  
##  0  1  2  3  4  5  6  7  8  9 10 11  
## 369 326 357 353 190 180 150 190 180 206 196 140
```

```
range(timeAlone) #range
```

```
## [1]  0 11
```

```
var(timeAlone) #variance
```

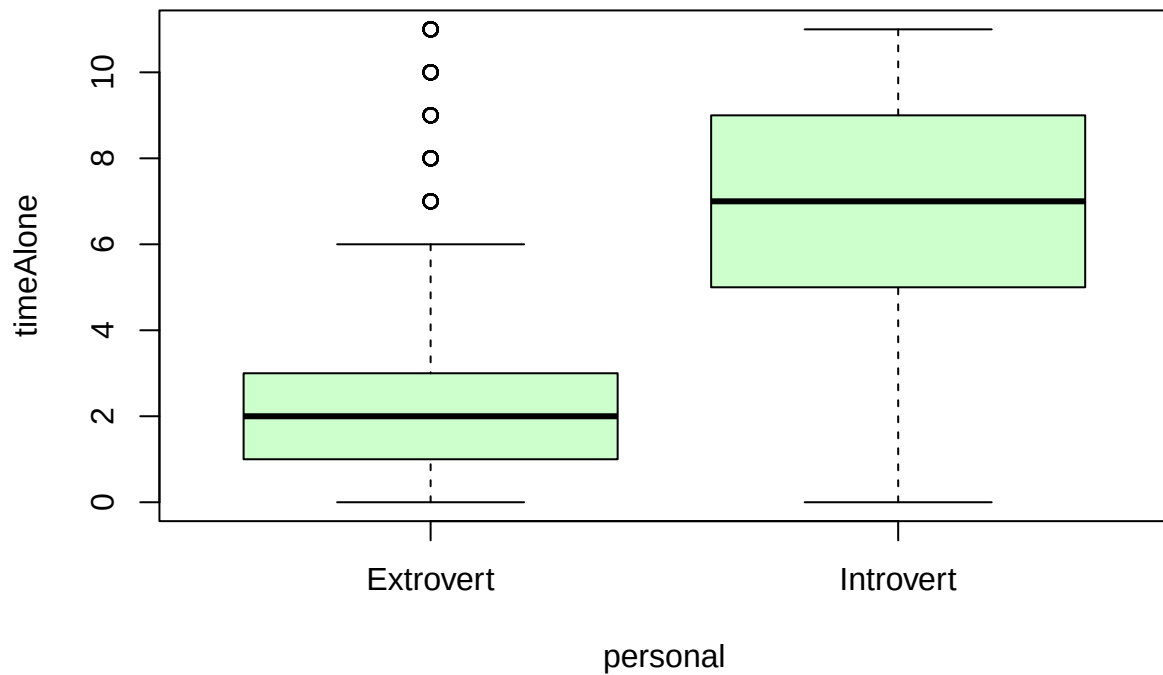
```
## [1] 12.10478
```

```
sd(timeAlone) #standard deviation
```

```
## [1] 3.479192
```

```
boxplot(timeAlone ~ personal, col = "#ccffcc", main = "Relation between personality and time alone") #b
```

Relation between personality and time alone



This was everything used in order to show the quantitative data.

Conclusion

Doing this activity made me realize that there are lots of programs that helps with the use of data and the way someone might use it to analice data.